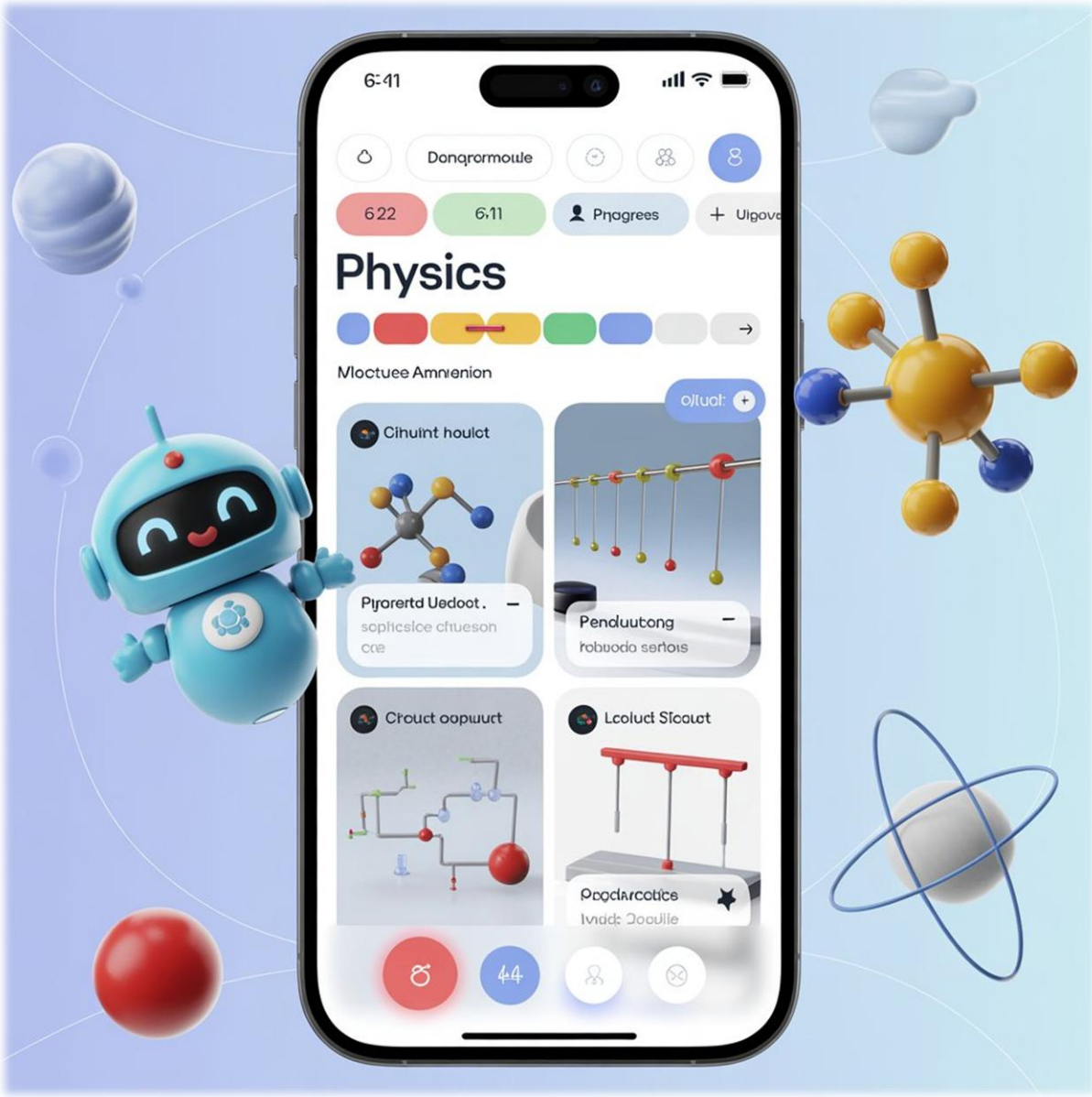
Mobile-Based Virtual Physics Laboratory with AI Assistance and Interactive Learning for Rural Students



R26-IT-150

Overall, Title and Content

Real-World Problem

- Physics education requires practical experiments to understand concepts effectively.
- Many rural schools in Sri Lanka lack laboratory facilities and equipment.
- Students mainly learn physics through theory without hands-on experience.
- Limited access to teachers and resources reduces understanding of complex concepts.
- Existing learning systems do not integrate experiments, quizzes, and progress tracking.
- Students do not receive personalized feedback to improve weak areas.

Scenario Demonstration (2-Minute Video)



Existing Solutions vs Proposed Solution

Existing Solutions

- Current educational systems provide virtual labs, quizzes, games, chatbots, and basic learning analytics separately.

Limitations of Existing Solutions

- Most systems focus only on one feature (experiments, quizzes, or chatbot) separately.
- AI chatbots are not adapted to student grade level or lesson context.
- Learning analytics systems do not combine theory and practical experiment data.
- Many platforms do not provide personalized feedback or learning recommendations.

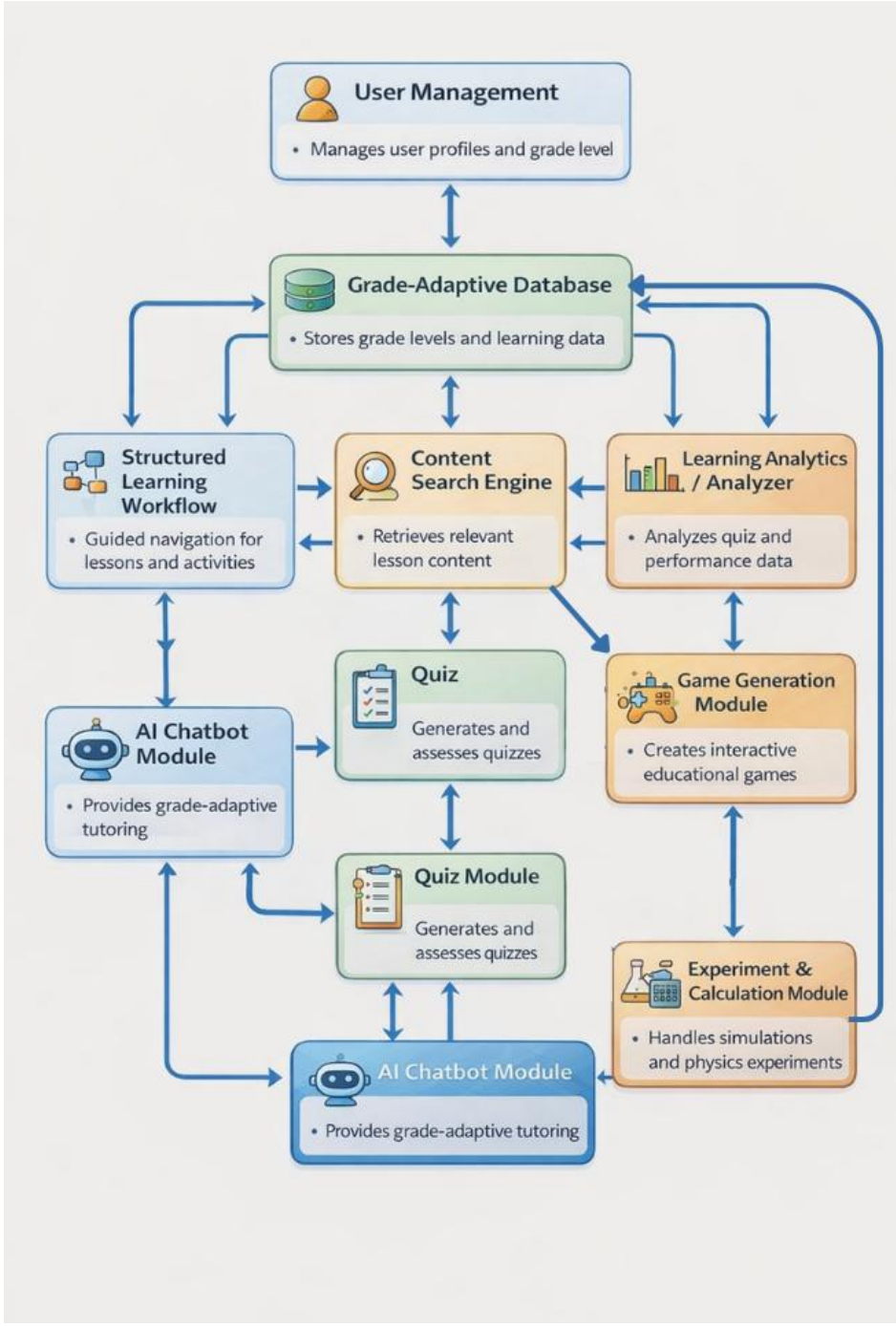
Proposed Solution

- **Mobile-based Virtual Physics Laboratory** with games, experiments, AI chatbot, and quizzes in one system.
- **Interactive simulations generate real-time physics measurements and step-by-step calculations.**
- **Accessible learning platform for Grade 6–11 students**, especially in schools with limited lab facilities.

How the Proposed System Overcomes Limitations

- **Integrates multiple learning features** (experiments, quizzes, games, AI support) in one platform.
- **Grade-adaptive AI explanations** based on lesson context.
- **Unified learning analytics** combining theory results and experiment performance.

System Architecture



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Information Technology



Design and Development of a Grade-Adaptive RAG-Based AI Chatbot Module and a Structured Learning Workflow Module

Contribution to Overall Solution:

- Design and Development of a Grade-Adaptive RAG-Based AI Chatbot Module and a Structured Learning Workflow Module

Novelty:

- Curriculum- and Workflow-Aware Grade-Adaptive RAG Tutor (CWGA RAG Tutor)

Grade-Adaptive RAG Based Chatbot and Learning Workflow Module

PROBLEM DEFINITION :

Develop an AI-based grade-adaptive chatbot and structured learning workflow system to support Grade 6–11 students with curriculum-aligned physics learning in a mobile virtual lab.



Research Objective:

Main Objective:

To design and develop a Curriculum- and Workflow-Aware Grade-Adaptive RAG-Based AI Chatbot Module and a Structured Learning Workflow Module for Grade 6–11 physics students

Sub – Objectives:

- Build a Grade-Tagged Physics Knowledge Base
- Implement a RAG-Based AI Chatbot Pipeline.
- Develop Grade-Adaptive Response Generation
- Design and Implement the Structured Learning Workflow Module
- Integrate the Chatbot with the Mobile Frontend
- Validate and Evaluate System Performance

Grade-Adaptive RAG Based Chatbot and Learning Workflow Module

Research Methodology (Design Science Research)

- Apply Design Science Research (DSR) to design and evaluate the AI chatbot and workflow system.
- Use physics syllabus content, lessons, and student questions as system inputs.
- Apply LLM and RAG methods for grade-adaptive response generation.
- Implement structured learning workflow for guided navigation.
- Evaluate using accuracy, usability, and student feedback.

Technologies Used

📱 **Flutter** – Used to integrate the AI chatbot interface and structured learning workflow screens into the main mobile application.

☁️ **Firebase Firestore** – Used to store curriculum content references, workflow metadata, chat history, and learning navigation data.

🐍 **Python Backend (FastAPI)** – Used to handle chatbot requests, workflow context processing, RAG retrieval, and communication between the mobile app and AI services

🧠 **Large Language Model (LLM)** – Used to generate physics explanations, question answering, and guided learning support for students.

🔎 **Retrieval-Augmented Generation (RAG)** – Used to retrieve relevant syllabus content

Grade-Adaptive RAG Based Chatbot and Learning Workflow Module

Validation and Evaluation

- **Chatbot Performance:** Accuracy, relevance, grade-appropriateness, and response consistency.
- **RAG Performance:** Retrieval accuracy, context relevance, and grounded response generation.
- **Workflow Validation:** Evaluate menu hierarchy, lesson navigation flow, and ease of access to learning content.
- **Model Validation:** Compare chatbot responses with curriculum content and expected grade-level explanations.
- **Usability Testing:** Student feedback, teacher feedback, and overall system usability within the mobile app.

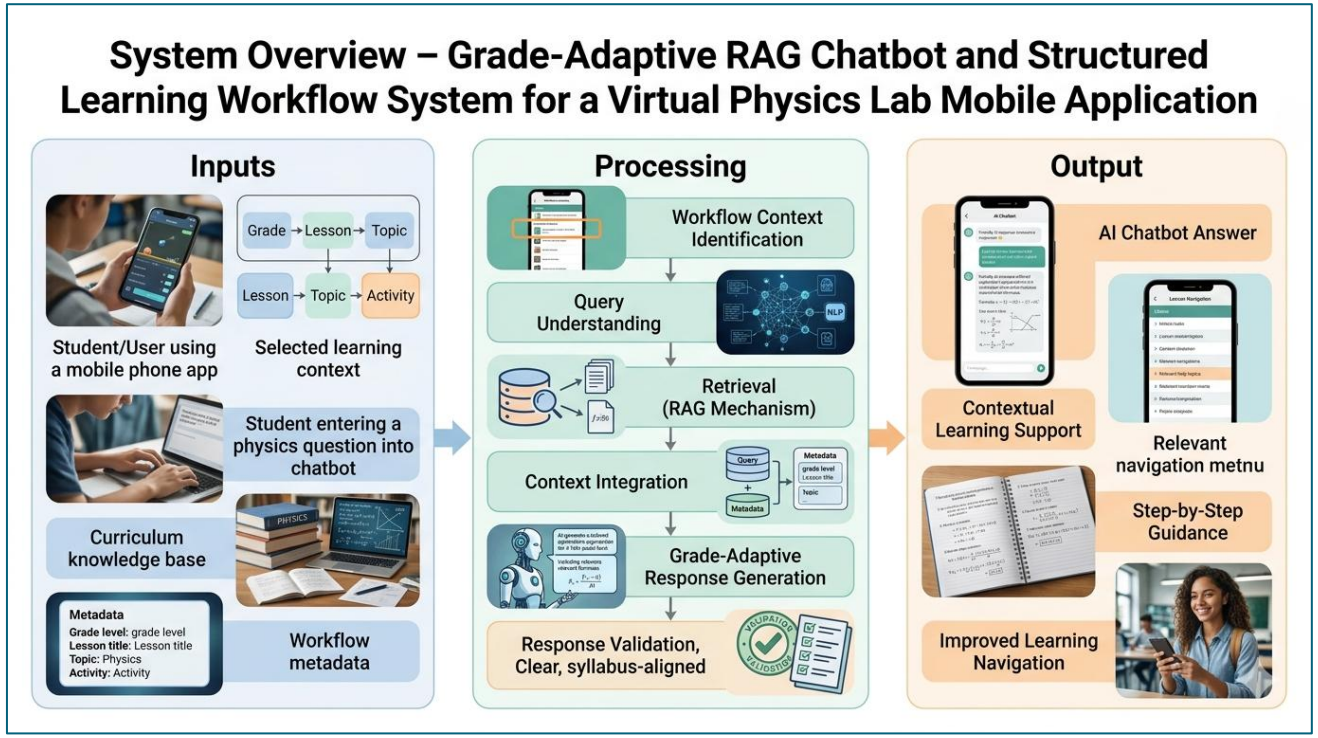
Data, Security and Ethical Considerations

- **Physics learning dataset** collected from syllabus materials, lesson notes, practical guides, and educational resources.
- **Student data and chat interactions** handled with consent and used only for research and system improvement purposes.
- **Chat records and workflow data** stored securely in protected databases or cloud storage.
- **Collected data** used only for educational support, research evaluation, and module development purposes.
- **AI responses** restricted to curriculum-related physics learning to reduce misleading or unsafe outputs.

Grade-Adaptive RAG Based Chatbot and Learning Workflow Module

Implementation Details

System Overview of the Individual Component



Dependencies and Common Components

- **Computer Vision Models –**
  optional OCR for text extraction
- **Deep Learning Frameworks –**
  LLM and RAG for chatbot responses
- **Backend API Services –**
  mobile and AI system communication
- **Shared System Components –**
  mobile UI, knowledge base, cloud storage

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Information Technology



Intelligent Keyword-Based Content Retrieval and Curriculum-Aligned 2D Game Generation Module.

Contribution to Overall Solution:

- My module provides keyword-based physics search and generates grade-appropriate 2D games to reinforce lessons for Grade 6-11 students

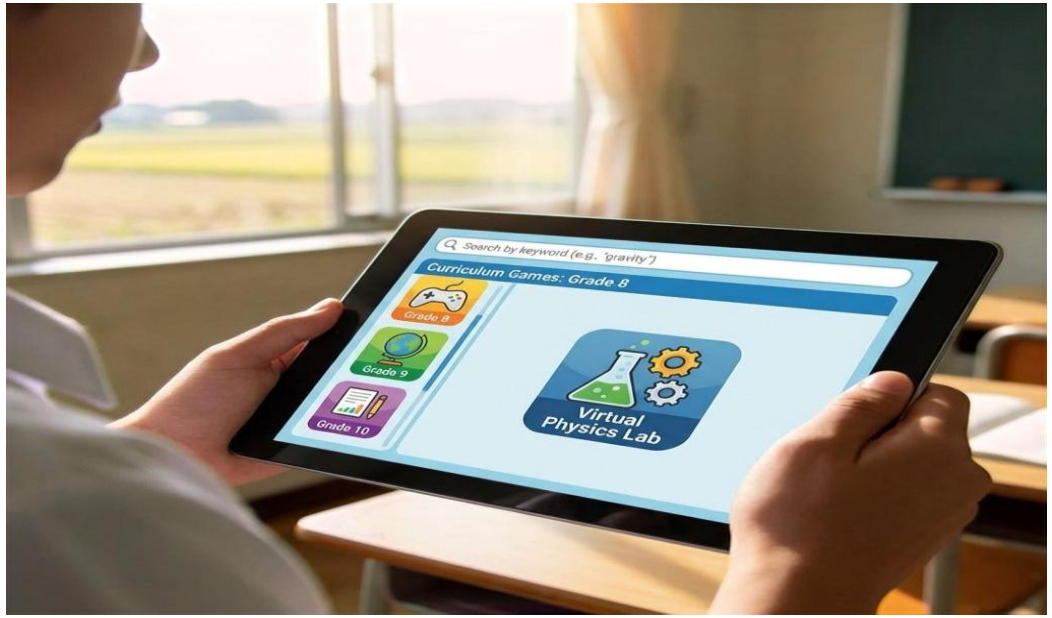
Novelty:

- A syllabus-aware framework that adapts physics search results and 2D gameplay difficulty to each student's specific grade.

Keyword-Based Physics Search and Adaptive Gaming Assistant

Problem Definition :

Rural students lack laboratory access and guided physics instruction. Current digital tools lack integrated, keyword-based search and curriculum-aligned games tailored to specific grade levels.



Research Objective:

Main Objective:

- Develop a keyword-based physics content retrieval system and curriculum-aligned 2D game generation module to enhance conceptual learning for rural students.

Sub – Objectives

- Implement a keyword-based search engine to retrieve relevant physics explanations and materials.
- Generate 2D educational games specifically aligned with national physics curriculum topics.
- Design a scoping framework to adapt game difficulty and content based on student grade level.
- Map digital learning activities to structured lesson-level objectives for Grades 6–11.

Keyword-Based Physics Search and Adaptive Gaming Assistant

Research Methodology (Design Science Research)

- Design and evaluate a **keyword-based search** and **curriculum-aligned gaming** system.

  Input: *Student-entered keywords and grade-level selection via mobile interface.*

- Apply **Intelligent Content Retrieval** to match keywords with Sri Lankan national physics syllabus materials.

- Generate **adaptive 2D educational games** that scope difficulty and content based on the student's grade level.

Technologies Used

- 📱 **Flutter / Android Studio**: Used for developing the lightweight mobile application for Android devices commonly used by students.
- 🎮 **Unity / Interactive 2D Tools**: Used to design virtual practical experiments and physics-based educational games.
- **Firebase Firestore:** Used as the central cloud database to store **NIE-aligned keywords**
- 💻 **Python**: Handles the backend application logic for the intelligent keyword retrieval and game generation module

Keyword-Based Physics Search and Adaptive Gaming Assistant

Validation and Evaluation

- **Search Performance**:
  Keyword accuracy and content retrieval relevance.
- **Educational Accuracy**:
  Alignment with Sri Lankan National Curriculum standards.
- **System Usability**:
  Student feedback and mobile interface usability testing.
- **Engagement Evaluation**:
  Learner participation and motivation in 2D gameplay.

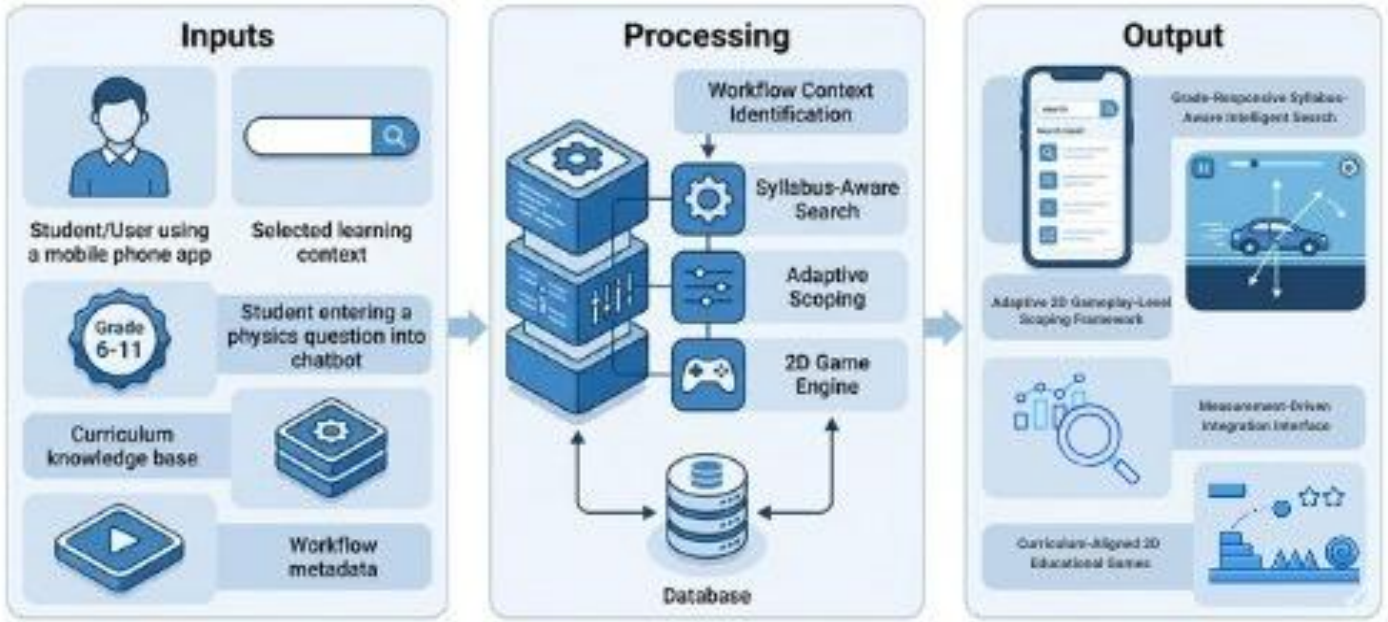
Data, Security and Ethical Considerations

- **Official Sourcing**: Physics content is extracted from official Sri Lankan textbooks and teacher guides to ensure syllabus accuracy.
- **User Privacy**: Student profiles and activity logs are protected through authenticated user management and secure login systems.
- **Data Security**: Quiz results and performance tracking data are stored securely within the unified student learning analytics module.
- **Ethical Scoping**: All search results and 2D games are strictly tailored to be age-appropriate for Grade 6–11 learners.
- **Resource Sensitivity**: The system is optimized as a lightweight mobile application to function in rural areas with limited technological infrastructure.

Keyword-Based Physics Search and Adaptive Gaming Assistant

Implementation Details

System Overview of the Individual Component



Dependencies and Common Components

Machine Learning Models – pest detection from plant images

2D Game Development Framework

Keyword-Based Retrieval Systems – Matches student search queries with curriculum-aligned physics content

Backend API Services – Facilitates real-time communication between the game engine and the physics calculation services

Shared System Components – Flutter mobile interface , Firebase cloud database for syllabus data , and the National Institute of Education (NIE) curriculum dataset

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Information Technology



Adaptive User Management and Learning Analytics with Performance Evaluation and Feedback Recommendation Module

Contribution to Overall Solution:

My module manages student accounts, analyzes quiz and experiment performance, determines mastery levels, and generates personalized learning feedback

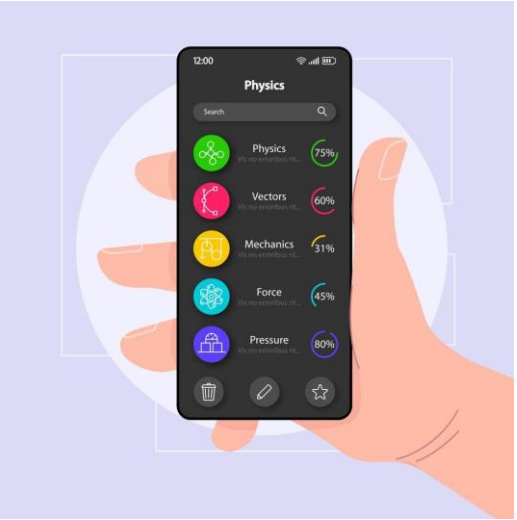
Novelty:

- This module integrates quiz results, virtual experiment performance, and lesson progression to analyze student learning, estimate mastery levels, detect weaknesses, and generate personalized feedback in a mobile virtual physics laboratory.

Adaptive User Management and Learning Analytics Module

Problem Definition :

Students often struggle to identify weak physics topics, while current learning systems rarely analyze theoretical quiz performance and practical experiment results separately, making it difficult to accurately evaluate mastery and provide targeted learning feedback.



Research Objective:

Main Objective: Develop an adaptive learning analytics module that analyzes quiz and experiment performance to identify weak topics, estimate mastery levels, and provide personalized feedback for physics learning.

Sub – Objectives

- Develop a secure student login and profile management system.
- Evaluate theoretical understanding using quiz performance.
- Analyze practical performance using virtual experiment results.
- Generate learning analytics and personalized feedback recommendations.

Adaptive User Management and Learning Analytics Module

Research Methodology (Design Science Research)

- Uses Design Science Research (DSR) methodology.
- Develop an adaptive learning analytics module for the mobile virtual physics laboratory.
- Input: Quiz results, experiment performance, and lesson completion data.
- Train a machine learning model using the ASSISTments dataset to analyze quiz performance patterns.
- Evaluate mastery level using quiz score, experiment score, and completion rate.
- Generate personalized feedback to improve weak physics topics.

Technologies Used

- **Flutter –** mobile application interface for students
- **Firebase Authentication –** secure student registration and login
- **Firebase Firestore –** storing quiz results, experiment data, and learning records
- **Machine Learning (Decision Tree) –** quiz performance analysis and mastery level prediction
- **Python Backend (FastAPI / Flask)**
- **ASSISTments Dataset –** training data for quiz performance pattern analysis

Adaptive User Management and Learning Analytics Module

Validation and Evaluation

Mastery Level Prediction Evaluation: Metrics: Accuracy, Precision, Recall, and F1-Score

- **Quiz Performance Analysis Evaluation:** Compare predicted mastery with actual quiz performance patterns
- **Learning Analytics Effectiveness:** Evaluate system ability to identify weak topics and student strengths
- **System Usability Evaluation:** Student feedback and usability testing to assess system usefulness and learning support

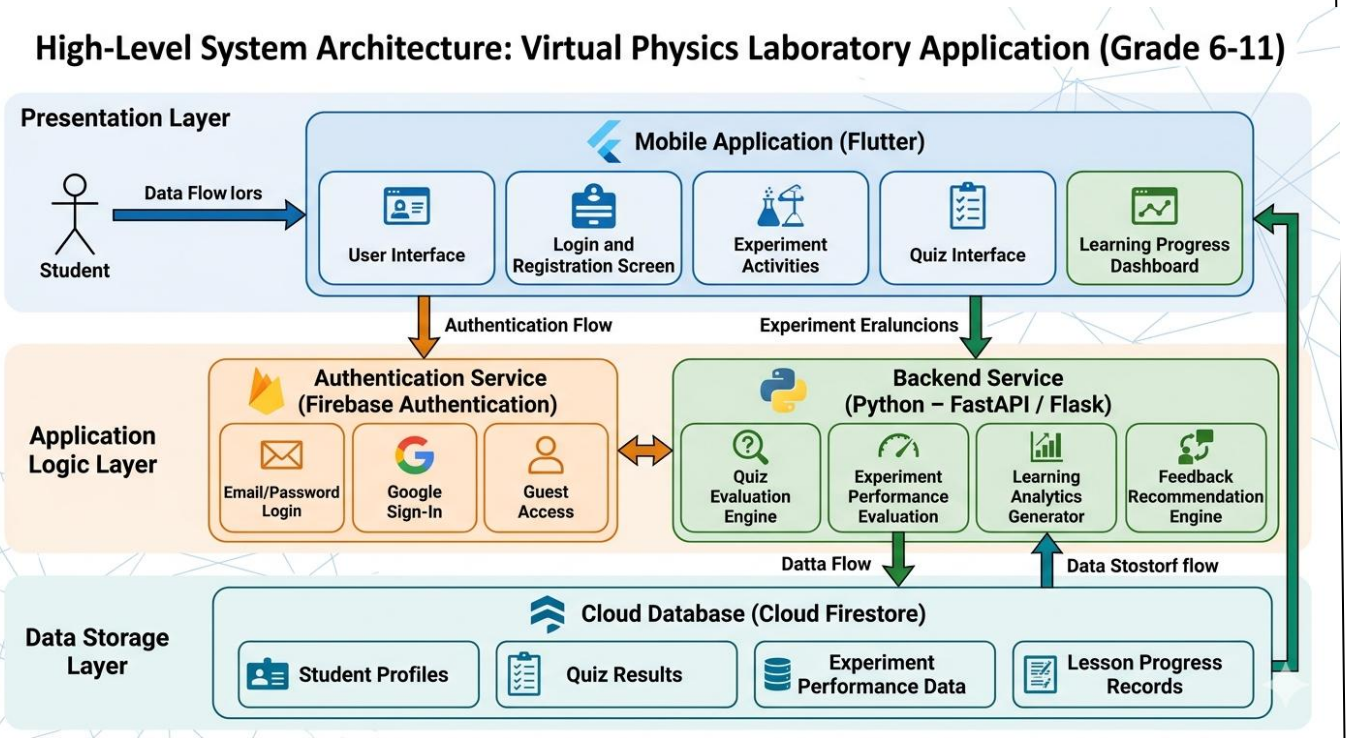
Data, Security and Ethical Considerations

- **ASSISTments dataset** used for training quiz performance analysis models.
- **Student quiz results, experiment scores, and learning progress data** collected through the mobile application.
- All student data used **only for research and system development purposes.**
- **User authentication and access control** used to protect student data privacy.
- Learning records and performance data **securely stored in cloud databases (Firebase).**

Adaptive User Management and Learning Analytics Module

Implementation Details

System Overview of the Individual Component



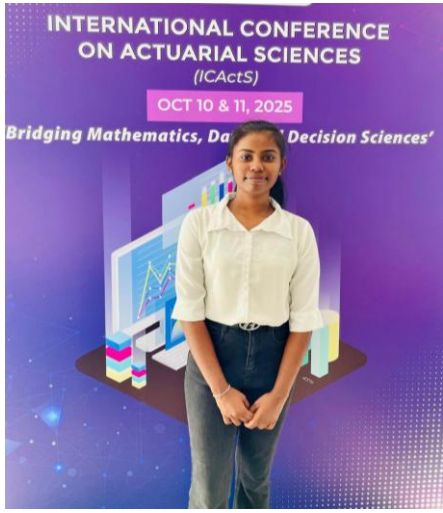
Dependencies and Common Components

- **Machine Learning Model** – quiz performance pattern analysis and mastery level prediction
- **Learning Analytics Module** – analyzing quiz results, experiment scores, and lesson completion
- **Backend API Services** – communication between mobile application and analytics services
- **Shared System Components** – mobile interface, user authentication, and cloud database for learning records

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Information Technology

Interactive Physics Game-Based Measurement, Calculation and Experimentation Learning Module



Contribution to Overall Solution:

- Enable interactive physics learning through games and virtual experiments with automated measurement-based calculations and evaluation.

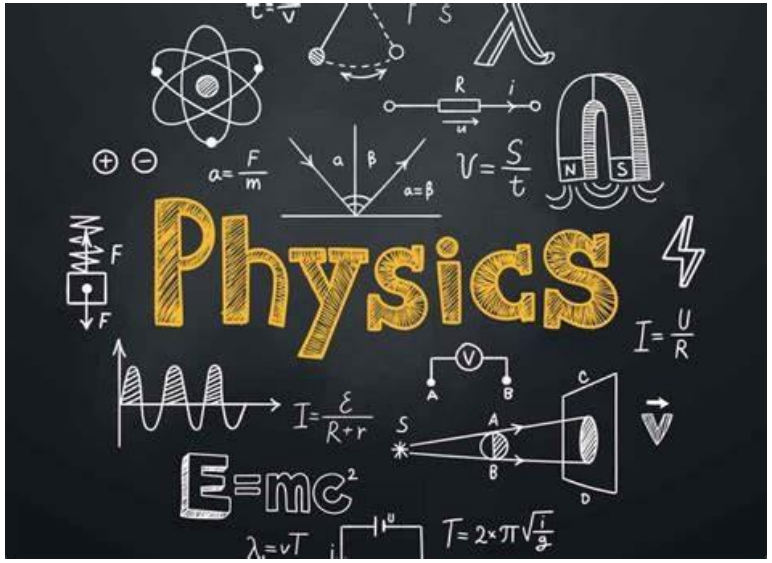
Novelty:

- Converts student interactions into real-time measurements and automatically explains step-by-step physics calculations for concept understanding.

Learning Physics Interactive Measurement and Virtual Experiments

Problem Definition :

Many students lack access to laboratory facilities and struggle to understand physics concepts through theory alone. This component addresses the need for interactive physics learning through virtual experiments and games that generate measurable results and structured experiment evaluation.



Research Objective:

Main Objective:

- Develop an interactive physics learning system that integrates physics-based games and virtual experiments to help Grade 6–11 students understand physics concepts through practical interaction.

Sub – Objectives

•Capture measurements .

• Apply physics formulas to generate calculation results.

• Create virtual laboratory experiments with digital tools.

•Evaluate experiment performance.

Learning Physics Interactive Measurement and Virtual Experiments

Research Methodology (Design Science Research)

- Develop physics games and virtual experiments.
- Capture measurements such as distance, time, and velocity.
- Extract physics measurements from simulation data.
- Apply physics formulas to generate results.
- Evaluate experiments based on steps, tools, and results.

Technologies Used

📱 **Unity 2D -** Create virtual experiments

🤖 **Pre Trained Models (OpenCV / YOLOv8) –** Object detection and Motion tracking

⚡ **Python –** backend services

📱 **Flutter –** mobile application

☁️ **Firebase –** cloud storage

Learning Physics Interactive Measurement and Virtual Experiments

Validation and Evaluation

The system will be evaluated using:

- **Measurement Accuracy:** Accuracy of generated physics measurements
- **Calculation Accuracy:** Correctness of physics formula calculations based on measured values.
- **Experiment Evaluation Accuracy:** Correct scoring of safety tools, apparatus selection, procedure steps, and final result.
- **System Usability:** Student feedback and usability testing of the interactive learning module

Data, Security and Ethical Considerations

- Physics measurements and experiment interaction data are generated from student activities within games and virtual experiments.

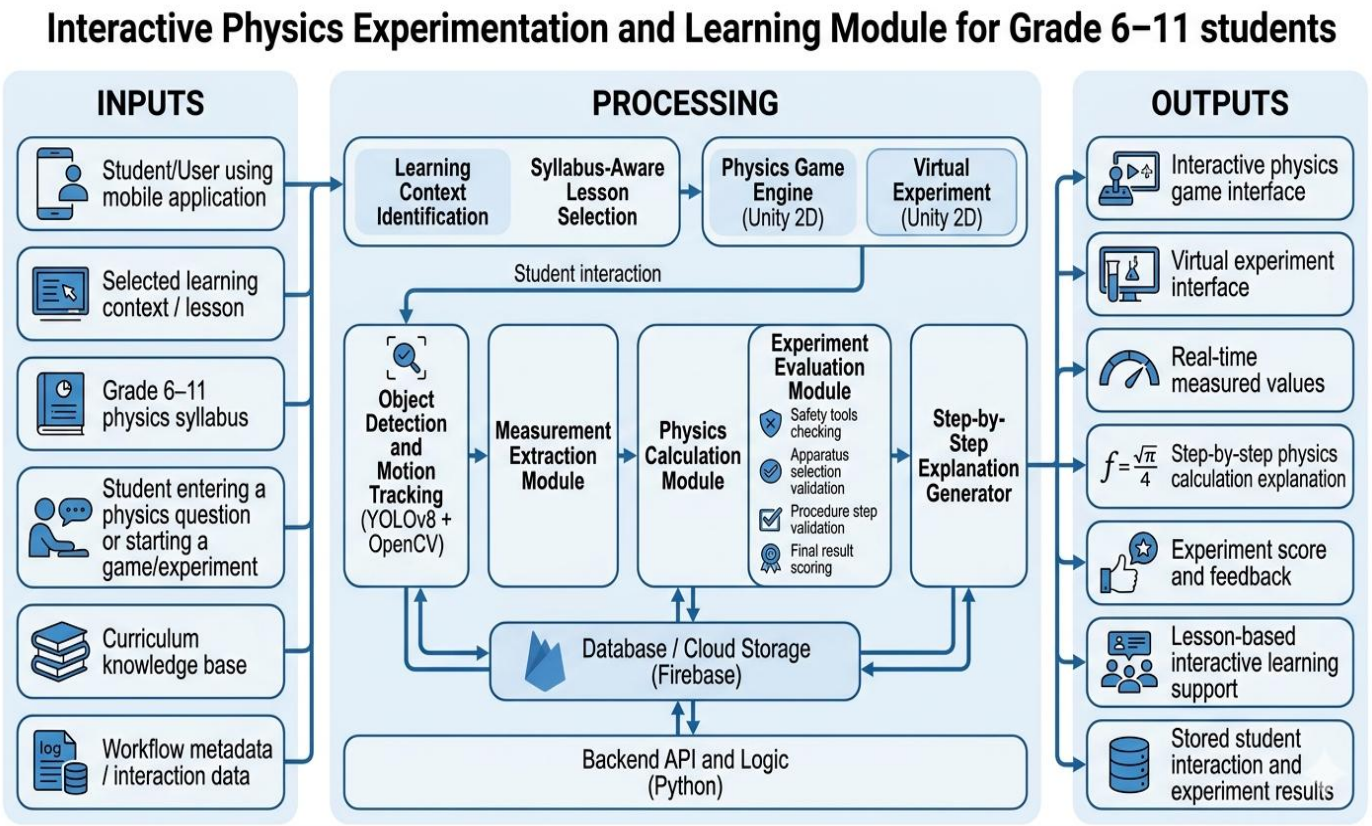
- Interaction data and experiment results stored securely in cloud database

- Data used **only for research purposes**

Learning Physics Interactive Measurement and Virtual Experiments

Implementation Details

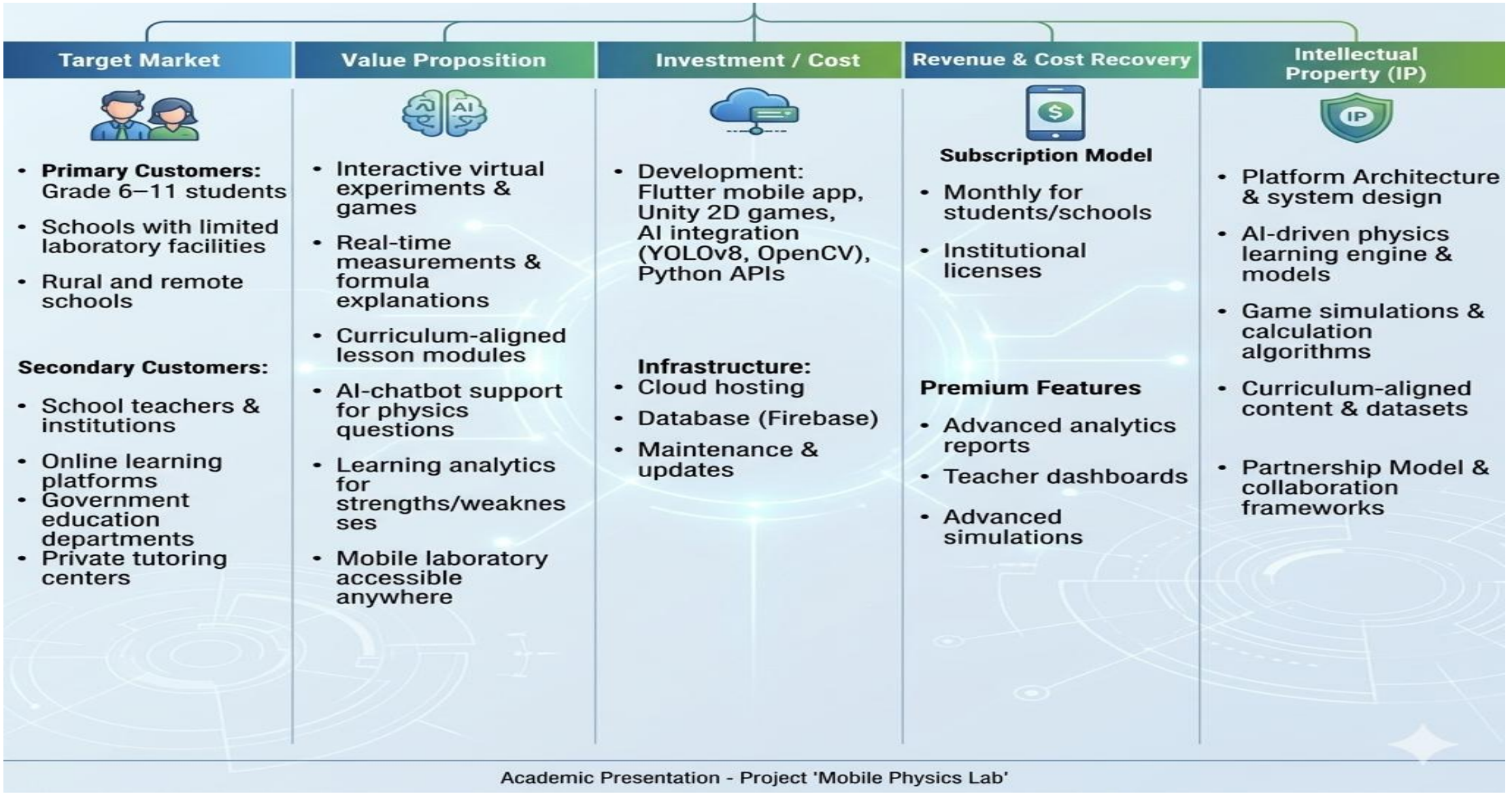
System Overview of the Individual Component



Dependencies and Common Components

- **Simulation Engine (Unity 2D)** – used to develop physics games and virtual experiments.
- **Physics Calculation Module** – applies physics formulas to measurements
- **Backend API Services** –manage communication between the mobile application and the calculation or data processing modules
- **Shared System Components** – mobile interface, cloud database, and experiment results and interaction data storage

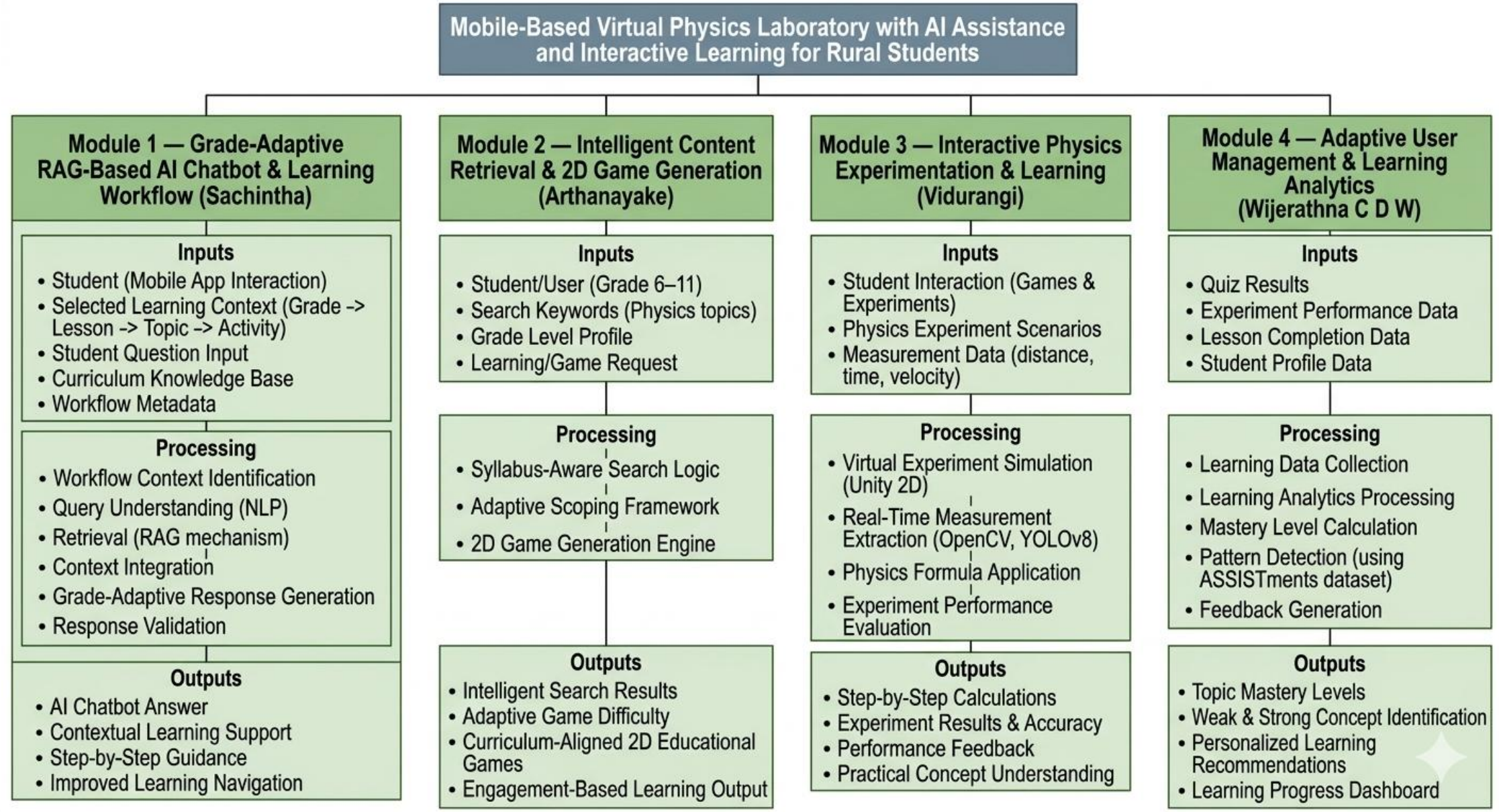
Business Model and Market Strategy



GANTT CHART

Task / Activity	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12
Topic Selection & Project Planning	■	■										
Literature Review	■	■	■									
Requirement Analysis		■	■									
System Architecture Design			■	■								
Database & UI Design			■	■								
User Management Module Development				■	■	■						
Learning Performance Evaluation Module				■	■	■						
AI Chatbot Development					■	■	■					
Virtual Experiment Development					■	■	■	■				
Gamified Learning Development					■	■	■	■				
System Integration							■	■				
System Testing & Debugging								■	■			
System Evaluation									■	■		
Documentation & Report Writing									■	■	■	
Final Submission & Presentation											■	■

WORK BREAKDOWN STRUCTURE



REFERENCES

Dataset Request Letter

THANK YOU !